

Eunah Jeong

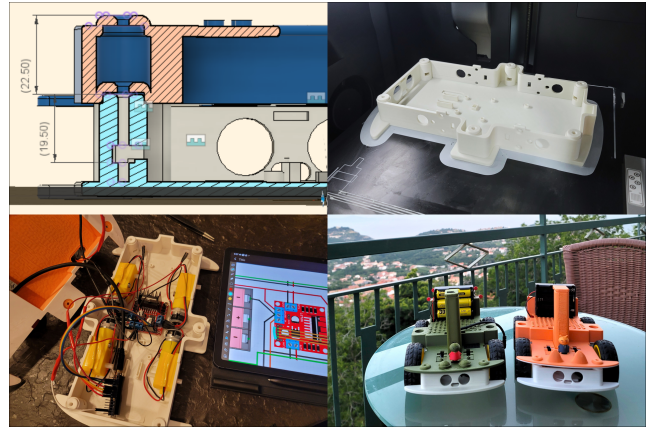
Motivation Letter & Research Statement

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Dear EPFL E3 Program Committee,

My motivation stems from an enduring curiosity to learn and to push the boundaries of both cognitive and physical capability. I have come to value learning as a deeply creative act through my Foundation Visual Arts certificate, 3D & Game Design diploma, experiences working as a designer, and exploration of diverse disciplines, from theoretical studies to embodied skills such as piano, sketching, sewing, and ballet. Yet I often found myself alone in this pursuit, surrounded by indifference toward curiosity and growth. This solitude led me to robotics, where I see the potential for intelligent, collaborative systems to share the joy of learning and creation that I sought among people.

Ultimately, my vision is to develop robots that not only act intelligently but also embody enthusiasm and creativity, enabled by the power of cooperation, transforming how we learn, work, and create together.



OpenBot Assembly. *I have modeled, 3D printed, built, and operated these low-cost robots.*

Two long-term, multifaceted projects were pivotal early learning experiences for me that revealed how important it is to have capable “hands” on a project to elevate the final outcome. These projects, in which I experienced the full process of video and game production, taught me invaluable lessons. They involved understanding lighting, audio, and cinematography; 3D modeling; environment building; UX/UI design; and physics simulation; culminating in the creation of [fashion show promotional videos](#) and a [game trailer](#).

Now, with a Computer Science B.Sc. and greater exposure to scientific research through my ongoing B.E. program, I have become particularly intrigued by how robots can be educated efficiently and affordably to collaborate across diverse tasks. My exposure to the manufacturing industry, where cost is a key factor in decision-making, further deepened my fascination with developing low-cost, scalable robot systems that learn from experience and from one another, enabling richer coordination than current one-to-one learning frameworks allow. To this end, I have worked on projects like building an [OpenBot Fleet](#) ([Müller et al., 2024](#)) and exploring [generalist policies](#) for low-cost robots. I am acutely aware of the limitations of current learning paradigms, with open-source [Vision-Language-Action \(VLA\) models](#) requiring at least 16 GB of GPU memory for inference, exceeding my 8 GB capacity. I have come to believe that a strong foundation in engineering and dedicated study of robot learning can help address such obstacles that limit creative endeavor—both my own and others’—and enrich the ways humans and machines learn and create together.

The vision of Prof. Aude Billard, Prof. Maryam Kamgarpour and their research groups at EPFL captivates my curiosity. Prof. Billard’s emphasis on bridging models and data, and on connecting AI and control theory, as discussed in [the ICRA 2025 debate](#) and [her interview](#) at University of Edinburgh resonates deeply

with me. Likewise, Prof. Kamgarpour's work on game and control theory has inspired my interest in multi-robot cooperation and learning; her research on constrained correlated equilibria in Markov games (Ni et al., 2025) sparked my thoughts on modeling robot coordination among multiple interacting robots with their own goals and constraints.

Attending the [CoRL 2025 Data Workshop](#) affirmed my research orientation: I am more drawn to approaches that integrate principled models with learning, rather than relying solely on large volumes of data to discover patterns. It also made me realize the importance of strengthening both my theoretical foundation and hands-on experimentation to build the skills necessary for advancing robotics. In this regard, I believe that the research environments led by Prof. Billard and Prof. Kamgarpour at EPFL, where rigorous theoretical insight meets practical implementation, offer the ideal setting for me to develop these capabilities and define my path as a researcher. By bringing my unique perspective as a ever-curious learner and learning enthusiast to EPFL's collaborative research community, I hope to contribute to the ongoing effort toward a society where humans and robots learn and coexist meaningfully.

Thank you for considering my application.

Eunah Jeong